MTH219

Timed Online Assignment – July Semester 2021

Fundamentals of Statistics and Probability

Thursday, 18 November 2021 **1:00 pm – 3:30 pm**

Time allowed: 2.5 hours

INSTRUCTIONS TO STUDENTS:

1. This Timed Online Assignment (TOA) contains **FOUR (4)** questions and comprises **SEVEN (7)** pages (including cover page).

2. You must answer **ALL** questions.

3. If you have any queries about a question, or believe there is an error in the question, briefly explain your understanding and assumptions about that question before attempting it.

4. You MUST **submit your answers via Canvas (similar to TMA submission) at the end time of this TOA** (as stated on this cover page). The 15 minutes grace period as shown on Canvas is strictly meant for technical issues encountered during submission. Thereafter, you will not be able to submit your answers and you will be considered as having withdrawn from the course. **No appeal will be allowed.**

5. You are to include the following particulars in your submission: TOA Course Code, your Full Name, and SUSS PI Number. Name your submission file as TOACourseCode_FullName_StudentPI. Use underscore and NOT Space. *Example: TOAXYZ_RaphaelLee_T1923161 (omit D/O, S/O).*

6. Your submission should consist of only one file and must not exceed 500MB in size. The file must be a PDF or Microsoft Word file saved in .doc or .docx format. All answers are to be typed. Flowcharts and graphs may be scanned or photographed and embedded in the PDF or Word file provided it does not exceed the file size limit of 500MB.

7. For answers which cannot be typed and require to be hand-written, ensure that question number is clearly stated on each page. All uploaded hand-written answers must be clear, readable and complete. Marks will not be awarded for un-readable or incomplete images.

8. To prevent plagiarism and collusion, your submission will be reviewed by Turnitin. The Turnitin report will only be made available to the marker and you will not be able to view it.

9. The University takes plagiarism and collusion seriously, and your Turnitin report will be examined thoroughly as part of the marking process.

10. Marks will only be awarded if FULL working is shown.

Answer all questions. (Total 100 marks)

Question 1

(a) x_i for $i = 1, \ldots, 51$ are the number of apples sold in a fruit store in 51 different days. You may assume that the number of apples sold for each day is independent from one another.

$$\sum_{i=1}^{51}(x_i - 70) = 676$$

$$\sum_{i=1}^{51}(x_i - 70)^2 = 23718$$

(i) Calculate the mean and variance of the dataset.

(5 marks)

(ii) Using the value of the variance calculated in Question 1(a)(i), calculate the value of $\sum_{i=1}^{51} x_i^2$.

(4 marks)

(b) Cindy throws a fair cubic die with faces numbered 1, 2, 3, 4, 5, 6. If she throws an odd number then her score is the number thrown. If she throws an even number, she will throw the die a second time and her score is the sum of both numbers drawn (Figure Q1(b)).

Let the random variable X denote Cindy's score.

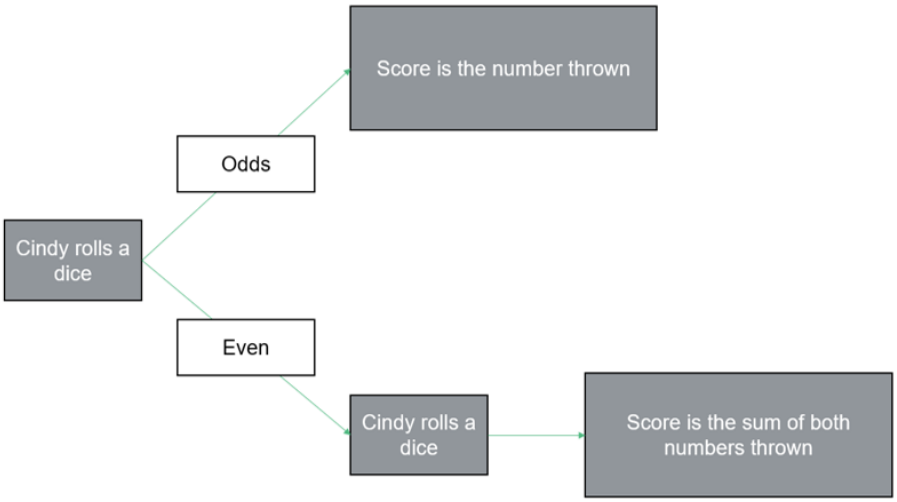

Figure Q1(b)

(i) Compute the frequency table for Cindy's score. You may find it helpful to create a table with all possible dice rolls and the scores.

x	1	2	3	4	5	6	7	8	9	10	11	12
Frequency, f												
Probability, p(x)												

(8 marks)

(ii) And hence computer $E(X)$ and $Var(X)$.

Hint:

$$E(X) = \sum x\, P(x) \;\; for\ all\ X$$

$$E(X^2) = \sum x^2\, P(x)\, for\ all\ X$$

(8 marks)

Question 2

Bag A contains 4 balls numbered 1, 3, 5, 7.
Bag B contains 5 balls numbered 5, 7, 7, 7, 9.
Bag C contains 7 balls numbered 1, 4, 6, 7, 7, 7, 7.

One ball is selected at random from each bag.

Event X is 'exactly two of the selected balls have the same number'.
Event Y is 'the ball selected from bag C has number 6'.

(a) (i) Compute the probability that 2 "Ones" will be drawn.

(2 marks)

(ii) Compute the probability that 2 "Fives" will be drawn.

(2 marks)

(iii) Compute the probability that 2 "Sevens" will be drawn.

(4 marks)

(iv) Hence or otherwise, compute $P(X)$.

(4 marks)

(b) Describe in words the event $X \cap Y$. What are all the possible scenarios?

(3 marks)

(c) Compute $P(X \cap Y)$ and hence determine whether events X and Y are independent.

(6 marks)

(d) Compute the probability that two balls are numbered 5, given that exactly two of the selected balls have the same number.

(4 marks)

Question 3

A business owner set up an online shop with an e-Commerce Site. Based on his previous observations, he found that items were purchased at the following rates:

Day/Night	Time	Rate
Day	9 am to 9 pm	20 per hour
Night	9 pm to 9 am	5 per hour

Table Q3

(a) It is proposed that instead of splitting the day into Day and Night, we can simplify the analysis by considering the purchase rate on a daily basis instead.

(i) What is the daily average rate? (Ignore time)

(2 marks)

(ii) Explain if the Poisson Distribution is an appropriate model to study the purchases over the entire day (24 hours, ignoring time).

(5 marks)

(b) Let X be the number of purchases in Day time (9 am to 9 pm). Poisson distribution may be assumed to be suitable model for X.

(i) Apply the Poisson Distribution and compute the probability for exactly 8 purchases in any half hour.

(3 marks)

(ii) Let T be the waiting time to the next purchase in the Day time (9 am to 9 pm). Write down the distribution of T.

(2 marks)

(iii) What is the expected value of T?

(2 marks)

(iv) Using the CDF of T or otherwise, calculate $P(T > 5)$, the probability that the waiting time to the next purchase exceeds 5 minutes.

(4 marks)

(c) We will now model the number of purchases at the night time. There are an average of 5 purchases per hour from 9 pm to 9 am. Assume that the Poisson distribution is a suitable model.

From 9 pm to 9 am, there are 12 "1 hour" periods, specifically:

1. 9 pm to 10 pm;
2. 10 pm to 11 pm;
…;
12. 8 am – 9 am

(i) What is the probability for a single "one hour" period to have at least 5 purchases?

(4 marks)

(ii) What is the probability that for all "1 hour" periods to have at least 5 purchases each?

(3 marks)

Question 4

(a) The probability density function of the random variable X is given by

$$f(x) = \begin{cases} \frac{2}{9}\, x\,(3-x) & 0 \le x \le 3, \\ 0 & Otherwise. \end{cases}$$

(i) Sketch the graph of $y = f(x)$ and state the median of X.

(4 marks)

(ii) A statistical analyst wishes to compute the probability of $X < 1.5$. With the help of the graph drawn in Question 4(a)(i), show how can he compute $P(X < 1.5)$.

(2 marks)

(iii) Given that $P(X > 0.5) = 0.9259$, compute the value of $P(0.5 < X < 1.5)$.

(2 marks)

(b) The time Caden spends on gaming each day has a normal distribution with mean 1.8 hours and standard deviation σ hours. On 90% of these days, he spends more than 1.3 hours on his gaming.

(i) Determine the value of σ.

(3 marks)

(ii) Compute the probability that, on a randomly chosen day, Caden spends less than 2 hours on gaming.

(3 marks)

(c) The MTH219 module assessment weighting in 2020 was

Assessment	Weight Allocation	Marks	
		Mean	Standard Deviation
Examination	70%	53.2	9.3
CMA	8%	85.6	15.7
TMA	16%	65.4	19.6
PCQ	6%	95.3	8.3

Table Q4(c)

(i) Calculate the mean and standard deviation of the overall marks (weighted average of the four assessments). Assume that the marks in each assessment are independent from one another.

(5 marks)

(ii) What is the proportion of students who are expected to score an A grade, if A grade is 75 marks and above?

(3 marks)

(iii) Do you think that the assumption "the scores from the 4 assessments are independent" is reasonable? State your reasoning clearly.

How will the mean and standard deviation of the overall marks be affected if the marks in each assessment are not independent from one another?

(3 marks)

----- END OF PAPER -----